

ELECTIVE COURSE-2.5

URA751: CYBER PHYSICAL SYSTEM				
	L	T	P	Cr
	2	0	2	3.0
Course Objective: Learn more the technologies and components needed for communicating an in-depth knowledge of Industry 4.0. The modular and flexible design has a range of learning scenarios, from individual pallet transfer systems with integrated controller right up to a connected production system with cloud services.				
Syllabus Production planning and control with MES4: controlling the CP lab in the default mode, the principles of production planning and control (PPC), Control and data management of the CP factory in the MES mode, analyzing and optimizing the CP lab, Introduction to Cyber physical system, Live AR, MES based production planning, Visualization of plant process, smart maintenance, implementation of Industry4.0 process, pallet transfer system, closed loop factory view. Physical realization of CP lab system: commissioning CP lab system, developing simple library-compatible blocks, developing a process controller for a CP Lab station, Integration of identification systems, Integration of MES system, starting factoryviews control panel, configuration of apps, Starting the factoryviews user interfaces, Starting MES4, terminating Factoryviews. Creating AR scenes with Live Data: Identifying the required control variables, configuring the middleware for querying and forwarding the value, configuring an appropriate element in the scene description file, testing the developed solution. Industrial communication with OPC-UA: Basics of communication in the industry 4.0 environment, introduction to OPC-UA, OPC-UA client server communication. Industrial communication with PROFINET: Basics of PROFINET and PROFIBUS, setting up a profinet based network, maintenance of profinet, adapt networking, troubleshoooting, ring topology. Energy efficiency in production: Fundamentals of energy efficiency, energy flows and				

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively. Further, revision Approved in 1st meeting of Academic Council held on September 11, 2025.

reference data, Basics of energy efficiency, energy efficiency measures in production.

Laboratory Work

1. Design and structure of the CP Lab
2. Recording information using intelligent sensors
3. Control using PLC
4. Communication based on bus technologies
5. Binary pallet identification
6. Identification via RFID
7. Plug & produce: quick integration of new application modules using cyber-physical systems
8. Manufacturing execution system (MES): creating, managing, controlling and visualizing customer orders

Course Learning Outcomes (CLOs)

The students will be able to:

1. understand the networking of different industrial applications and the production plan using MES4.
2. compare the ERP systems and MES, the two most common systems for production control, as well as their tasks and characteristics.
3. create products in Web shop (Pricing I Groups I Filters), preparing product titles, product details as well as the product presentation for the online sales channel.
4. customize the product details (availability, delivery time) in the online sales channel.

Text Books

1. Principles of Cyber Physical Systems, Rajeev Alur, MIT Press, 2015
2. E. A. Lee, Sanjit Seshia , "Introduction to Embedded Systems – A Cyber–Physical Systems Approach", Second Edition, MIT Press, 2017, ISBN: 978-0-262-53381-2

Reference Books & Websites

1. Plant control and Commissioning with MES, LX
2. Industrial communication with OPC-UA, LX
3. Industrial communication with PROFINET, LX
4. Energy efficiency in production, LX
5. Lx.festo.com

Hardware Reference:

- 1) CP Lab 404-1

Evaluation Scheme:

Sl. No	Evaluation Elements	Weightage (%)
1	MST	35
2	EST	35

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively. Further, revision Approved in 1st meeting of Academic Council held on September 11, 2025.